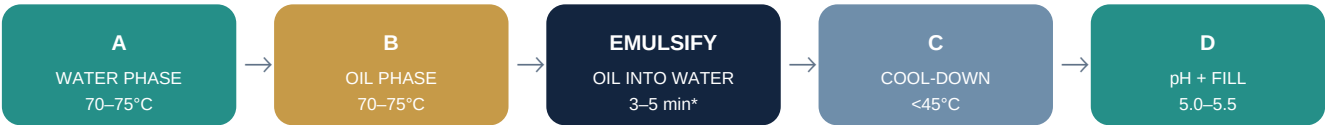


DAY CREAM

FORMULATION & MANUFACTURING SPECIFICATION

Final approved master formula • Manufacturer R&D: sample document

A lightweight facial day cream designed for hydration, soft skin feel, fresh-looking glow, reduced appearance of dullness and a premium, fast-spreading sensory experience.



Document control	Value
Document type	R&D; formulation specification
Formula revision	v3 — supplied master formula
Product name	DAY CREAM
Product type	Facial Day Cream
Document date	02 October 2026
Formula data hash	19cdf27ecb3
Status	For laboratory R&D; sample; not commercial release

FINAL MASTER FORMULA — DO NOT MODIFY WITHOUT APPROVAL

This document preserves the supplied formulation. The Phase D line is retained as supplied: a 0.30% process pool consisting of 10% lactic-acid solution as required for pH adjustment and water make-up as required. The actual split must be recorded in the batch record.

1. Product information

Field	Specification
Intended use	Daily leave-on facial moisturiser for morning use. Apply to clean skin; sunscreen remains recommended for daytime use.
Product concept	Premium lightweight O/W facial day cream with a restrained active system and a soft, fast-spreading finish.
Target skin types	Normal, dry, combination and oily skin as reasonably realistic; fragrance-free variant recommended for sensitive users.
Desired sensory profile	Silky spread, quick absorption, hydrated but not wet, soft after-feel, low tack, low grease, no waxy drag and subtle fragrance.
Intended cosmetic benefits	Hydration, softness, smoothness, fresh-looking glow, reduced appearance of dullness and support for more even-looking skin tone.
Fragrance	0.15% facial fragrance as supplied; supplier must provide applicable IFRA documentation and allergen declaration.
Packaging recommendation	Opaque airless pump, preferably 30–50 g consumer pack; minimise finger contact, oxygen ingress and light exposure.
Claims guardrail	Cosmetic appearance claims only. No medical acne, pigmentation, melasma, whitening or guaranteed four-day claims.

Product performance target

Immediate performance should be driven primarily by water-binding ingredients, light emollients and the emulsion structure. Longer-term tone support is gradual and customer-dependent. The formula is not a sunscreen and does not replace sunscreen.

2. FINAL MASTER FORMULA

FINAL MASTER FORMULA — DO NOT MODIFY WITHOUT APPROVAL

The table below is the controlled master data extracted from the supplied v3 formula. Total = 100.00% and 1,000.00 g at 1 kg scale.

Raw Material	INCI Name	% w/w	100 g	1 kg	Phase	Function
Deionised water	Aqua	54.58	54.58 g	545.80 g	A	Solvent
Disodium EDTA	Disodium EDTA	0.10	0.10 g	1.00 g	A	Chelator
Glycerin	Glycerin	4.00	4.00 g	40.00 g	A	Humectant
Xanthan gum	Xanthan Gum	0.20	0.20 g	2.00 g	A	Rheology / stabiliser
Niacinamide	Niacinamide	4.00	4.00 g	40.00 g	A	Barrier / tone / sebum support
Allantoin	Allantoin	0.20	0.20 g	2.00 g	A	Soothing / smoothing
Olivem 1000	Cetearyl Olivatate, Sorbitan Olivatate	4.00	4.00 g	40.00 g	B	O/W emulsifier
Cetearyl alcohol	Cetearyl Alcohol	0.50	0.50 g	5.00 g	B	Co-emulsifier / body
Caprylic/capric triglyceride	Caprylic/Capric Triglyceride	4.00	4.00 g	40.00 g	B	Light emollient
Squalane	Squalane	2.00	2.00 g	20.00 g	B	Light emollient
Tocopherol	Tocopherol	0.20	0.20 g	2.00 g	B	Oil antioxidant
Deionised water (cool-down)	Aqua	20.00	20.00 g	200.00 g	C	Solvent for cool-down actives
Sodium hyaluronate	Sodium Hyaluronate	0.10	0.10 g	1.00 g	C	Humectant / film former
Alpha-arbutin (COA, HQ n.d.)	Alpha-Arbutin	1.50	1.50 g	15.00 g	C	Tone-evening active
Zinc PCA	Zinc PCA	0.50	0.50 g	5.00 g	C	Sebum / NMF support
D-panthenol 75%	Panthenol (per supplier INCI)	2.67	2.67 g	26.70 g	C	Humectant / soothing; 2.00% active target
PE 9010	Phenoxyethanol, Ethylhexylglycerin	1.00	1.00 g	10.00 g	C	Preservative
Facial fragrance (IFRA)	Parfum	0.15	0.15 g	1.50 g	C	Sensory
10% lactic acid + water process pool	Lactic Acid, Aqua	0.30	0.30 g	3.00 g	D	pH q.s. / make-up
TOTAL		100.00	100.00 g	1,000.00 g		

Phase totals: A 63.08% / 630.80 g; B 10.70% / 107.00 g; C 25.92% / 259.20 g; D 0.30% / 3.00 g process pool.

3. Raw-material specifications

The following requirements are formulation-development controls. Where the supplied formula does not provide a numerical specification, the requirement is explicitly left for supplier or manufacturer confirmation.

Raw material	Grade / COA / SDS / storage / handling requirement
Deionised water	Purified water suitable for cosmetic manufacture. Microbial quality, conductivity and storage/hold time: TO BE CONFIRMED WITH SUPPLIER/MANUFACTURER. Use closed, clean storage.
Disodium EDTA	Cosmetic grade. Assay, microbial quality, COA and SDS required. Keep dry and sealed. Exact supplier specification: TO BE CONFIRMED WITH SUPPLIER/MANUFACTURER.
Glycerin	Vegetable cosmetic grade; IP/USP-type documentation acceptable if appropriate. COA, SDS and storage information required. Hygroscopic; keep tightly closed.
Xanthan gum	Cosmetic-grade clear/soft grade preferred. COA, SDS and hydration information required. Keep dry; disperse in glycerin to avoid clumping.
Niacinamide	Cosmetic-grade or suitable pharmaceutical-grade raw material with COA, assay, microbial and SDS documentation. Nicotinic-acid-related quality requirement: TO BE CONFIRMED WITH SUPPLIER/MANUFACTURER. Keep dry and sealed.
Allantoin	Cosmetic grade with assay, COA and SDS. Keep dry. Confirm solubility and supplier handling information.
Olivem 1000	Supplier-equivalent Cetearyl Olivat/Sorbitan Olivat material. TDS, COA, SDS, melting/processing information and identity confirmation required. Keep cool, dry and sealed.
Cetearyl alcohol	Cosmetic grade with identity, assay/quality and SDS. Keep dry. Melting range and supplier specification: TO BE CONFIRMED WITH SUPPLIER/MANUFACTURER.
CCT	Cosmetic grade Caprylic/Capric Triglyceride. COA should include supplier quality parameters applicable to the material; SDS required. Store closed, cool and protected from excessive heat/light.
Squalane	Cosmetic-grade squalane, not squalene. Source and quality documentation required. Protect from excessive heat, light and oxidation.
Tocopherol	Cosmetic-grade tocopherol; active form/content and COA required. Store tightly closed, cool and protected from light. Not a preservative.
Sodium hyaluronate	Cosmetic grade; molecular-weight range, assay/quality, COA and SDS required. Keep dry, sealed and protected from heat. Hydrate slowly.
Alpha-arbutin	Cosmetic grade. COA must include assay/purity and hydroquinone-related impurity specification. Storage, light/heat sensitivity and SDS required. HQ result must be documented by supplier; do not infer it.
Zinc PCA	Cosmetic grade with assay, COA, SDS and solubility/pH information. Keep sealed and dry. Avoid unapproved pH or electrolyte changes.
D-panthenol 75%	Cosmetic grade, 75% strength confirmed by COA. Supplier must confirm the complete INCI of the supplied liquid, storage and SDS. Formula quantity depends on 75% strength.
PE 9010	Preservative blend with exact composition, use range, COA, SDS and supplier TDS. Add below the supplied process temperature. Do not treat tocopherol as preservation.
Facial fragrance	Leave-on facial cosmetic grade; IFRA documentation and allergen declaration required. Storage closed, cool and dark. Do not use candle/soap fragrance.
10% lactic acid + water pool	Use a documented 10% lactic-acid solution or prepare under controlled conditions. Actual acid solution quantity is q.s. to pH; unused pool is water make-up. Exact supplier and pH-adjustment specification: TO BE CONFIRMED WITH SUPPLIER/MANUFACTURER.

4. Manufacturing SOP

The following procedure preserves the supplied manufacturing sequence and quantities. Equipment rpm must be set and recorded by the manufacturer for its specific laboratory homogenizer.

STEP 1 — Raw-material preparation

What to do: Verify raw-material identity, lot, COA and SDS. Sanitize vessel, utensils and filling components. Use a calibrated scale, calibrated thermometer and calibrated pH meter. Weigh materials by phase. Reserve Phase C water for the cool-down premix.

Target: clean, dry equipment. **Avoid:** unrecorded substitutions.

PHASE A — WATER PHASE

Material	100 g	1 kg	Order / condition
Deionised water	54.58 g	545.80 g	Charge first
Disodium EDTA	0.10 g	1.00 g	Add and dissolve
Glycerin	4.00 g	40.00 g	Use to slurry xanthan
Xanthan gum	0.20 g	2.00 g	Disperse into glycerin
Niacinamide	4.00 g	40.00 g	Add to water phase
Allantoin	0.20 g	2.00 g	Add to water phase

Procedure: Mix glycerin and xanthan into a smooth slurry. Charge Phase A water. Add EDTA and mix until dissolved. Add niacinamide and mix approximately 1 minute. Add allantoin and mix. Add the glycerin/xanthan slurry under mixing. Heat to 70–75°C and hold approximately 15 minutes with gentle mixing. **Expected:** uniform slightly viscous aqueous phase. **Do not:** add xanthan directly to water.

PHASE B — OIL PHASE

Material	100 g	1 kg	Order / condition
CCT	4.00 g	40.00 g	Charge liquid oil
Squalane	2.00 g	20.00 g	Charge liquid oil
Olivem 1000	4.00 g	40.00 g	Add flakes
Cetearyl alcohol	0.50 g	5.00 g	Add solid
Tocopherol	0.20 g	2.00 g	Add for final minute

Procedure: Combine CCT, squalane, Olivem 1000 and cetearyl alcohol. Heat to 70–75°C with mixing until all flakes are melted. Add tocopherol during the final minute. **Expected:** homogeneous melted oil phase with no visible flakes.

EMULSIFICATION

Bring Phases A and B within approximately 3–5°C of one another. Add Phase B to Phase A over approximately 20–30 seconds for the laboratory sample. Homogenize immediately with the manufacturer’s rotor-stator equipment for approximately 2–3 minutes as the starting laboratory condition, using a speed that avoids air entrainment. Record actual rpm, head and time. Follow with gentle sweep mixing during cooling. **Expected:** opaque, uniform cream with no oil beads or watery layer.

5. Cool-down, preservation, fragrance and pH

COOL-DOWN PHASE — PHASE C

Material	100 g	1 kg	Addition condition
Deionised water	20.00 g	200.00 g	Hydrate HA at room temperature
Sodium hyaluronate	0.10 g	1.00 g	Sprinkle slowly into water
Alpha-arbutin	1.50 g	15.00 g	Dissolve below 45°C
Zinc PCA	0.50 g	5.00 g	Dissolve below 45°C
D-panthenol 75%	2.67 g	26.70 g	Add below 40°C
PE 9010	1.00 g	10.00 g	Add below 40°C
Facial fragrance	0.15 g	1.50 g	Add below 35°C

Procedure: Cool the emulsion under gentle mixing to below 45°C. In the full Phase C water quantity, sprinkle sodium hyaluronate slowly and allow approximately 20–30 minutes hydration. Warm the premix to 35–40°C if required. Add alpha-arbutin and Zinc PCA to the same premix and mix until uniform. Do not exceed 45°C. Add the complete premix to the emulsion below 45°C and mix 1–2 minutes. Below 40°C add D-panthenol 75% and mix approximately 1 minute.

PRESERVATIVE

Below 40°C, add exactly 1.00 g per 100 g or 10.00 g per 1 kg PE 9010. Mix for approximately 2 minutes, ensuring complete distribution. Do not assume preservation without microbial and preservative-efficacy testing.

FRAGRANCE

Below 35°C, add exactly 0.15 g per 100 g or 1.50 g per 1 kg approved facial fragrance. Mix gently for 1–2 minutes. For the requested fragrance-free development version, prepare a separate controlled variant; do not silently change the master batch.

pH ADJUSTMENT — PHASE D

Target: pH 5.0–5.5 at approximately 25°C. Calibrate the pH meter with appropriate buffers. Measure after the batch has equilibrated near 25°C. Add 10% lactic-acid solution in small increments, mix approximately 60 seconds, and remeasure. The supplied formula allocates a maximum 0.30% process pool. If the full acid pool is not used, use water for the balance. Record the actual acid and water quantities. If pH is below 5.0, use an appropriately dilute pH-raising solution approved by the manufacturer. **FORMULA ITEM REQUIRES CONFIRMATION:** exact Phase D acid/water split must be established and recorded during the first R&D; batch; do not assume the full 0.30 g is lactic-acid solution.

FINALIZATION

Mix gently for 2–3 minutes below 35°C. Perform a final weight check and adjust to exactly 100.00 g or 1,000.00 g using purified water as required by the approved batch record. Check appearance, colour, odour, pH and absence of grit or visible separation. Fill below approximately 30°C into a sanitized opaque airless pump, minimizing headspace. Rest 24–48 hours before judging final texture and viscosity.

6. Manufacturing precautions

DO NOT:

- Change raw materials, suppliers or grades without approval and documentation.
- Change percentages or batch quantities without revising the controlled master formula.
- Add alpha-arbutin, sodium hyaluronate, preservative or fragrance to an excessively hot batch.
- Exceed 75°C during the hot-phase process without documented development approval.
- Add xanthan directly to water without pre-dispersion.
- Use less than the stated preservative amount without documented formulation approval.
- Treat tocopherol or fragrance as a preservative.
- Use essential oils, candle fragrance or soap fragrance in place of the approved facial fragrance.
- Change the pH target or release a batch outside pH 5.0–5.5 without documented approval.
- Assume that an attractive appearance proves microbiological safety or commercial shelf life.
- Release the sample without recording actual weights, temperatures, mixing times and deviations.

7. Finished-product specification

Parameter	Specification / target	When evaluated
Appearance	As established during R&D; smooth, homogeneous cream with no visible separation	Finished batch
Colour	As established during R&D;	Finished batch and stability
Odour	As established during R&D; and approved fragrance profile	Finished batch and stability
pH	5.0–5.5 at approximately 25°C	Finished batch and stability
Texture	Smooth, homogeneous cream	Finished batch
Viscosity	To be established during R&D; do not invent a release number before measurement	24–48 h and stability
Emulsion	No visible separation, creaming or syneresis	Finished batch and stability
Microbial quality	Laboratory specification	Finished batch
Stability	Must pass the approved stability program	Development gate
Preservative efficacy	Must pass the selected preservative-challenge method	Development gate
Active stability	To be established where applicable, including alpha-arbutin-related stability review	Stability

8. Testing requirements

The following tests are required to establish commercial suitability. Visual home or bench observation is not a substitute for qualified laboratory testing.

Test	Purpose
Appearance / colour / odour	Detect visible or sensory drift.
pH	Confirm the formula remains in the specified compatibility range.
Viscosity	Establish product identity and monitor drift; numerical limit to be set after R&D.;
Centrifuge	Screen for separation and emulsion weakness.
Freeze/thaw	Evaluate physical robustness across temperature cycling.
Accelerated stability	Assess heat-related changes and probable shelf-life risks.
40°C stability	Important for Indian climate and transport stress.
Packaging compatibility	Confirm no interaction with airless pump, actuator, piston, seals or pack materials.
Microbial testing	Confirm finished-product microbial quality.
Preservative challenge testing	Demonstrate preservation performance; required before commercial release.
Active stability	Evaluate relevant active retention and degradation, including alpha-arbutin-related impurity review where applicable.

Suggested observations: initial, 24 hours, 7 days, 14 days, 28 days and longer manufacturer-defined stability intervals. Test room temperature, 40°C, 4°C and freeze/thaw conditions as appropriate to the laboratory protocol.

9. Requested R&D; sample

REQUESTED DEVELOPMENT:

Prepare an initial laboratory R&D; sample using the exact FINAL MASTER FORMULA in this document. Prepare the fragranced master version and, preferably, a separate fragrance-free version as a controlled development variant.

The manufacturer must record: batch number; date; raw-material lot numbers; actual weights; phase weights; manufacturing temperatures; mixing and homogenization times; equipment and rpm; pH; final batch weight; final appearance; viscosity or texture observations; packaging used; and every deviation from the master formula.

10. Product performance and timeline

Time point	Realistic cosmetic expectation
Immediate cosmetic effect	Comfortable hydration, softer feel, smoother spreading, temporary fresh-looking glow and reduced tightness.
2–4 days	More consistent softness and a fresher-looking surface for some users; no guaranteed pigment or acne result.
1 week	Improved comfort and surface smoothness may be more noticeable with regular use.
2–4 weeks	Possible improvement in the appearance of dullness and overall skin feel; results vary.
6–12 weeks	Most realistic assessment window for gradual support of more even-looking tone. Daily sunscreen is important for tone-related goals.

These are cosmetic expectations, not guarantees. The formula is not an acne medicine, pigmentation treatment, sunscreen or medical product.

11. Packaging recommendation

Use an opaque airless pump where commercially practical. This is suitable because the formula is a water-based leave-on cream containing light- and heat-sensitive development materials, a preservative system and a low fragrance load. Airless packaging reduces finger contact, reduces repeated environmental exposure and supports a premium dosing experience. Confirm pump compatibility, dose consistency, piston movement, seal integrity and long-term pack interaction during development.

12. Regulatory and commercial note

This document is an R&D; formulation specification. It does not by itself establish commercial safety, stability, regulatory compliance or shelf life. Appropriate professional testing and regulatory review must be completed before commercial sale.

No regulatory approval, safety guarantee or commercial shelf-life claim is made by this document.

13. Formula change control

NO INGREDIENT, RAW-MATERIAL GRADE, PERCENTAGE, PROCESS TEMPERATURE, pH TARGET OR MANUFACTURING STEP SHOULD BE CHANGED WITHOUT DOCUMENTING AND APPROVING THE CHANGE.

Any proposed change must identify the original value, proposed value, reason, risk assessment, responsible approver, new batch number and required repeat testing. A manufacturer may recommend a change during R&D;, but the change is not part of this master formula until approved in writing.

14. Approval and issue record

Field	Entry
Product	DAY CREAM
Formula revision	v3 — supplied master formula
Formula total	100.00% / 1,000.00 g
Prepared for	Cosmetic manufacturer / chemist / R&D; laboratory
Status	Laboratory R&D; sample specification
Formula data hash	19cdaf27ecb3
Approval signature	_____
Manufacturer acknowledgement	_____
Date	_____

END OF CONTROLLED DOCUMENT